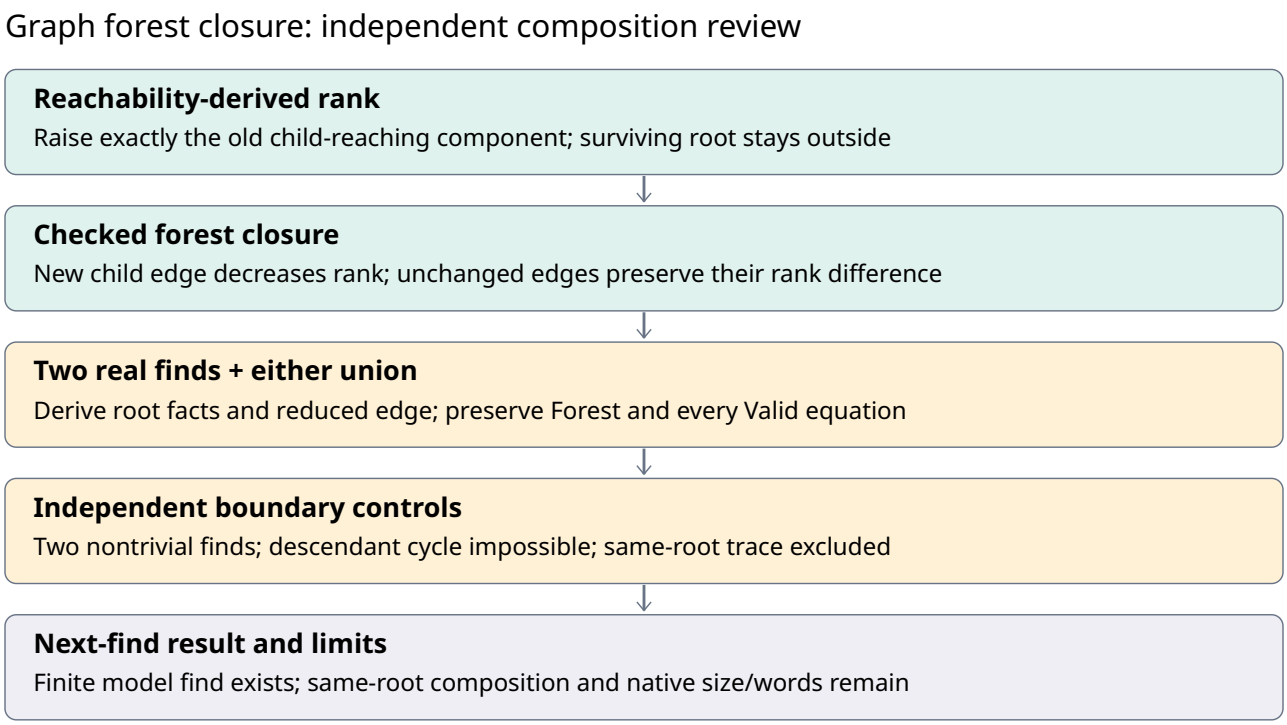
Independent review of graph forest closure



pencil-graph-forest-review-20261101 | 10 main + 9 producer + 15 reviewer | no science

Ten main declarations, nine producer controls and fifteen reviewer controls passed fresh replay, with thirty-four exact named axiom readbacks. The proof constructs an updated decreasing-rank forest witness and composes two actual find executions with either distinct-root union orientation. The review validates that component scope: the post-union forest and the post-find root facts are derived, not assumed.

Publication and independence. Sources were extracted from pushed PR3056 commit09e64fca55, storage-forest-adapter/graph-forest, and matched the producer's private bytes. All1599 producer manifest entries matched. The published coordinator review joint is copied beside this report. The reviewer has prior native SignedGraph source and find/union/cycle proof familiarity and authored the older GraphAlgebra import; this is independent review of the new forest/composition packet, not fully blind history. The local PLAN is an execution note after initial proof inspection, not a substitute for the published prospective plan. Exact current native source hashes are checked separately.

Constructed forest witness. Reach follows the actual parent function, with reflexivity and a parent-step constructor. root_reach proves a root cannot reach a distinct vertex. For x different from the child, reach_parent proves that x reaches the child iff its parent does. bumped adds old_rank(parent)+1 exactly to the vertices whose old path reaches the child. The surviving distinct root is outside that set. At the newly attached child, the new rank is strictly greater than the parent's unchanged rank. Every other strict edge has the same increment at both ends, or no increment at either; self-parent edges remain self-parented. This proves attach_forest from the original Forest. No poststate rank or acyclicity premise is supplied.

Root and component semantics. attachment_roots proves the actual child parent pointer and surviving root pointer. component_transferred derives a new path to the parent from each old path to the child, including descendants. The parent-root and distinctness premises exclude attaching a root into its own descendant chain. The attachment lemma does not require the child itself to be a root: cutting a nonroot subtree and attaching it to another distinct root can safely preserve a forest. Conversely, a nonroot proposed parent is outside the theorem's sufficient condition, not universally unsafe in all possible graphs.

Two-find composition. find_preserves_roots follows each recursive compression and shows an existing self-parent node cannot be overwritten as an active nonroot child. two_finds applies the established preservation theorem twice, carrying the same semantic vertex values and fixed anchors and retaining the first returned root after the second mutation. edge_after_two_finds then derives the reduced edge coefficients from the original edge and returned potentials, establishes the union relation and unit-square sign, and invokes branch_preserves. Both Boolean orientations yield

an existential constructed rank and the global Valid invariant. next_find_exists obtains a successful finite Run from that derived witness; it does not assume a successful future trace.

Independent controls. A five-node rational fixture has a signed path3->2->0 and a separate path4->1, with nonzero scalar offsets and an anchor at0. At k=2, vertex values are3,7,4,6,9. The first find compresses a two-edge path and the second find performs a separate nontrivial recursive step. Their true edge6+9=15 feeds both union orientations, and each result supports find existence for every vertex. Additional controls verify that the first root survives the second find, a descendant path transfers to the surviving root, and the two original roots cannot reach one another. Redirecting root0 to descendant3 produces a three-edge cycle and contradicts every possible Nat rank; the proposed parent is explicitly not a root.

Same-root boundary. Another checked second-find trace queries node2 after the first compression and returns root0 again. The distinct-root hypothesis is then false, so edge_after_two_finds does not apply. Native three_leads.rs1672-1679 takes its separate cycle/anchor branch when the roots agree. The reviewed composition covers the source's distinct-root branch1681-1711. Earlier cycle/anchor components remain separate; the new packet does not silently claim all same-root return behavior is composed.

Machine verification. The final fresh invocation compiled Find, Cycles, Union, Forest, Composition, PriorControls, ForestControls, Readback, Reviewer and ReviewerReadback at native/gate0, with unchanged bindings. accept.py requires those exact ten modules, exact source bytes and all34 named readback lines with only propext, Classical.choice and Quot.sound. A deliberate False theorem was rejected at native1/gate1 with an unsolved False goal. The first reviewer replay passed; a later fresh replay includes the additional same-component trace. Producer failed attempts remain preserved in the original packet. Cached dependency binaries and the Lean compiler are trusted; no independent binary kernel rechecker was run.

Concrete scope. The input assumptions remain an initial forest/Valid state, actual two Run witnesses, a true edge, distinct returned roots and unit-square decoded sign products. Run existence is available from the initial forest, but concrete word/sign decoding and compiled execution are separate. The bumped rank is a mathematical termination witness, not native size metadata or a performance estimate. Native union-by-size chooses between the two proved orientations; correctness/overflow of size arithmetic, node allocation, Vec bounds and counters are not established here. Same-root composition and full runtime bindings remain separate. No scientific execution, cloud action, repository edit or producer-snapshot mutation occurred.

Review outcome. Machine status passed; source fidelity faithful_scoped; review outcome validated_scoped for derived forest preservation, root/component semantics, two-find distinct-root union composition, global Valid preservation and subsequent model find existence. The diagram separates this result from the remaining native metadata and same-root interfaces.